Week 4 - Part 2 - Osprey1st College Algebra - Summer 2024

Section covered: 4.1 and 4.2

Rational Expression

A *rational number* is a number that can be written as a quotient of integers. A **rational expression** is also a quotient; it is a quotient of polynomials.

Rational expression
A rational expression is an expression that can be written in the form

$$\frac{P}{Q}, \text{ where } P \text{ and } Q \text{ are polynomials and } Q \neq 0.$$

Examples of Rational Expressions:

$$\frac{2}{9}, \quad \frac{3y-1}{4}, \quad \frac{x^2+1}{x-8}, \quad \frac{2y+5}{4y^3-y-8}$$

Excluded values
The values for which the rational expression is undefined are those that make the denominator 0 (since we cannot divide by 0).

For example, for the expression

$$\frac{x+10}{x-3}$$

the excluded value is 3, since $x = 3$ produced $\dfrac{13}{0}$.

Problem 1. Find all excluded values for the expression. That is, find all values of x, for which the expression is undefined. If there is more than one value, separate them with commas.

(a) $\dfrac{2x-10}{x}$ $\boxed{x \neq 0}$

(b) * $\dfrac{5x}{6+2x}$

$\cancel{6} + 2x = 0$
$-6 \qquad -6$
$\frac{\cancel{2}x}{\cancel{2}} = \frac{-6}{2}$
$x \neq -3$

(c) $\dfrac{6(x-1)}{x^2-5x-14}$

(c) $x^2 - 5x - 14 = 0$

-14; $+2$, -7; -5

$(x+2)(x-7) = 0$

$x+2=0 \qquad x-7=0$

$\boxed{x \neq -2 \quad x \neq 7}$

Equivalent Rational Expressions

$$-\frac{3}{8} = \frac{-3}{8} = \frac{3}{-8}$$

$$-\frac{5x+1}{3} = \underbrace{\frac{-(5x+1)}{3}}_{\text{Notice the parentheses}} = \frac{5x+1}{-3}$$

Simplifying Rational Expression To simplify a rational expression, or to write it in lowest terms, we use a method similar to simplifying a fraction.

We will simplify, for $b, c \neq 0$

$$\frac{a \cdot c}{b \cdot c} = \frac{a}{b} \cdot \frac{c}{c} = \frac{a}{b} \qquad \text{since} \qquad \frac{c}{c} = 1$$

Simplifying or Writing a Rational expression in Lowest terms

Step 1. Completely factor the numerator and denominator of the rational expression.
Step 2. Divide out factors common to the numerator and denominator.

Problem 2. Simplify each rational expression in lowest terms, if possible.

(a) $\dfrac{2x}{2x-12}$ $\dfrac{x}{x-6}$

(b) * $\dfrac{2u+7}{7u+2}$

(c) * $\dfrac{-2+v}{v-2}$

(d) * $\dfrac{-u-5}{u+5}$

(e) * $\dfrac{32w^4(w+4)(3w+2)}{40w^3(w+4)^3}$

(f) * $\dfrac{4x^2 - 4x - 80}{x^2 + 8x + 16}$

(g) * $\dfrac{4w^2 - 3}{w^2 + 6w - 7}$

Problem 3. Which of the following are not true? Explain why.

(a) $\dfrac{3-1}{3+5}$ simplifies to $-\dfrac{1}{5}$

(b) $\dfrac{4x-8}{4}$ simplifies to $x - 2$

(c) $\dfrac{2x+7}{2}$ simplifies to $x + 7$

(d) $\dfrac{8x+3}{8x}$ simplifies to $1 + 3 = 4$

Multiplying and Dividing Rational Expressions

Multiplication and Division

For fractions $\dfrac{a}{b}$ and $\dfrac{c}{d}$ ($b \neq 0$, $d \neq 0$), the following hold:

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \qquad \text{and} \qquad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}\,(c \neq 0).$$

Problem 4. Multiply or divide rational expressions as indicated.

(a) * $\dfrac{12y^3x}{x^5} \cdot \dfrac{3x}{4y^2}$

(b) $\frac{3y}{3-x} \cdot \frac{x^2-9}{12xy}$

(c) * $\frac{x-2}{3x-3} \cdot \frac{x^2+x-2}{x^2-5x+6}$

(d) * $\frac{b}{4a^2} \div \frac{5b^4}{6a^5b}$

(e) * $\frac{x^2+3x+2}{x-3} \div \frac{3x^2-12}{4x-12}$

(f) * $\frac{5x^2+11x+2}{5x+15} \div \frac{3x^2-20x-7}{x+3}$

Add or Subtract Rational Expressions

If $\frac{a}{c}$ and $\frac{b}{c}$ are rational expressions, then

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c} \quad \text{and} \quad \frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}$$

To add or subtract rational expressions with different denominators, find their least common denominator (LCD) and change each fraction to one with the LCD as denominator.

Finding the Least Common Denominator (LCD) or **Least Common Multiple**

1. Write each denominator as a product of prime factors.
2. Form a product of all the different prime factors. Each factor should have as exponent the greatest exponent that appears on that factor.

Problem 5. Find the LCD for each pair.

(a) $20x$ and $8x^2$

(b) * $16x^8v^6u^4$ and $8v^5u^7$

(c) $x(x+2)^2$ and $x^3(x+2)(x+1))$

(d) $x^2 - 25$ and $x^2 - 7x + 10$

Problem 6. Write each rational expression as an equivalent rational expression with the given denominator.

(a) $\dfrac{4}{5} = \dfrac{\quad}{35}$

(b) * $\dfrac{9}{4x^8} = \dfrac{\quad}{12x^9}$

(c) $\dfrac{7}{2x+5} = \dfrac{\quad}{6x^2+15x}$

(d) * $\dfrac{8y}{y-1} = \dfrac{\quad}{(y-1)(y-9)}$

Problem 7. Add or subtract rational expressions as indicated.

(a) * $\dfrac{x}{x^2+4x-12} - \dfrac{2}{x^2+4x-12}$

(b) * $\dfrac{2}{x-6} + \dfrac{3}{5x}$

(c) * $\dfrac{-9}{8x^2y} - \dfrac{7}{6xy^3}$

(d) * $\dfrac{x}{x-2} + \dfrac{1}{4x-8}$

(e) * $\dfrac{3}{x^2-7x-8} - \dfrac{2}{x+1}$

(f) * $\dfrac{2}{3x^2+2x-1} + \dfrac{1}{3x^2-4x+1}$

Complex Fractions

The quotient of two rational expressions is a **complex fraction**. There are two methods for simplifying a complex fraction:

Method 1. Multiply both numerator and denominator by the LCD of all the fractions.

Method 2. Add fractions in the numerator and the denominator separately by converting them to LCD. Then divide the fractions, that is multiply the numerator by the reciprocal of the denominator.

Problem 8. Simplify each complex fraction.

(a) * $\dfrac{\dfrac{x}{x+8}}{\dfrac{3x}{x^2+16x+64}}$

(b) * $\dfrac{\dfrac{4}{9x}+\dfrac{4}{3x^2}}{\dfrac{2}{3}+\dfrac{2}{x}}$

(c) * $\dfrac{1-\dfrac{3}{x+6}}{x+\dfrac{9}{x+6}}$

(d) * $\dfrac{\dfrac{5}{y+9}-\dfrac{1}{y-3}}{\dfrac{2}{y+9}+\dfrac{2}{y-3}}$